

Section 1.3: Truth Tables

Finite Mathematics MTH 107 Fall 2014

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Mars Hill University

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2014

/home/jim/JamesAshe/MHU/images/MarsHillU

Finding the Truth Value: Negation

Example 1.3.1.

Let p represent “I am an Elephant”. Find the truth value of p .

Finding the Truth Value: Negation

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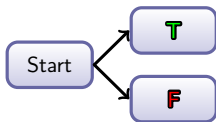
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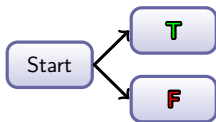


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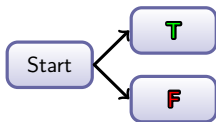


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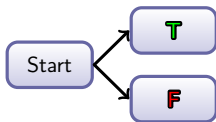
p	$\sim p$
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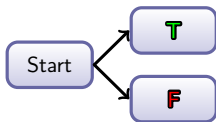
p	$\sim p$
T	

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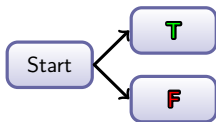
p	$\sim p$
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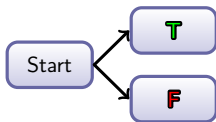
p	$\sim p$
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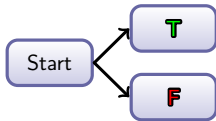
p	$\sim p$
T	F
F	T

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Example 1.3.1.

Let p represent “I am an Elephant”. Find the truth value of p .

Using a *truth table*.



p	$\sim p$
T	F
F	T

- Determine which statements are true and false, and choose the corresponding answer.

Finding the Truth Value: Conjunction, i.e., “and”

Example 1.3.2.

Let p represent “I am a man” and q represent “I am an Elephant”.
Find the truth value of $p \wedge q$.

Finding the Truth Value: Conjunction, i.e., “and”

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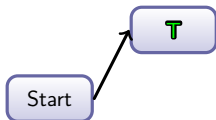
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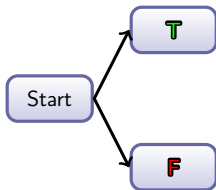


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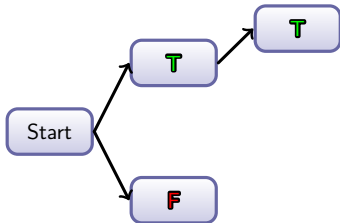


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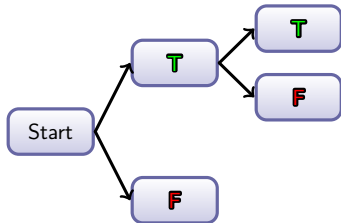


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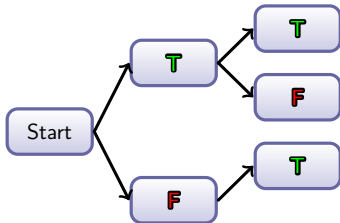


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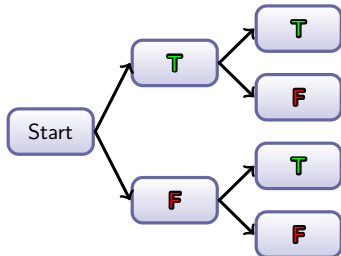


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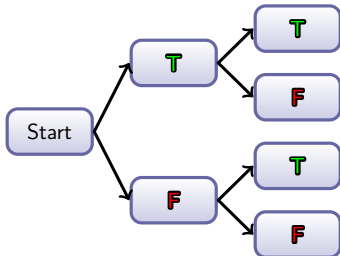


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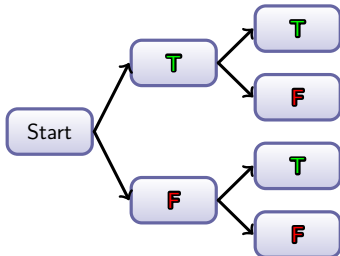
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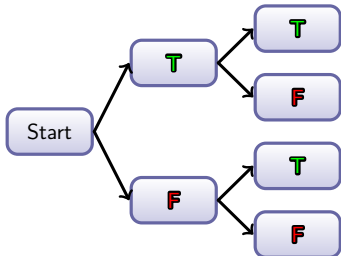
p	q	$p \wedge q$
T		
T		

Finding the Truth Value: Conjunction, i.e., “and”

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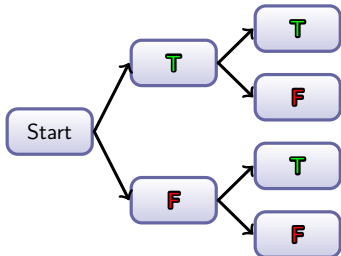
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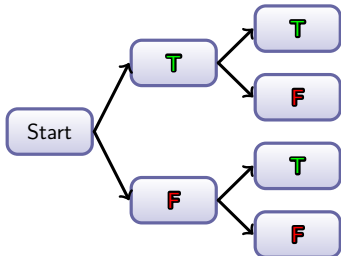
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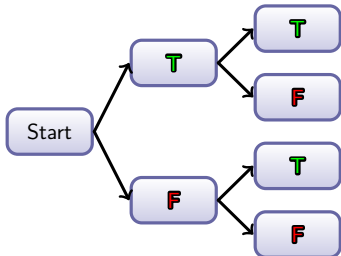
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p	q	$p \wedge q$
T	T	
T	F	
F	T	
F	F	

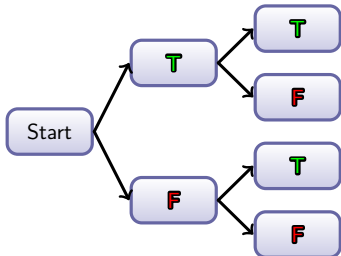
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T	T	T
T	F	
F	T	
F	F	

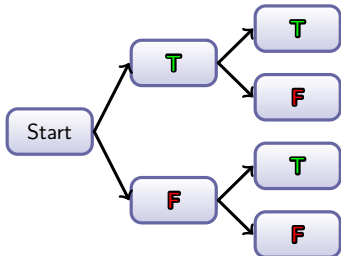
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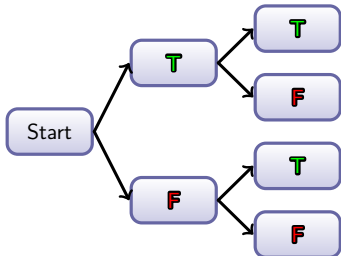
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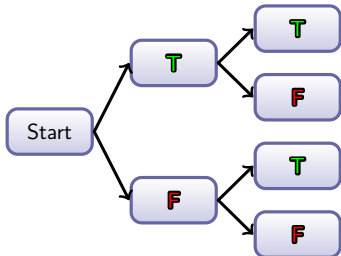
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F	T	F
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Finding the Truth Value: Conjunction, i.e., “and”

Example 1.3.3.

Let p be true and q be true. Find the truth value of $p \wedge q$.

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Using a *truth table*.

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Finding the Truth Value: Disjunction, i.e., “or”

Example 1.3.4.

Let p represent “I am a man” and q represent “I am an Elephant”.
Find the truth value of $p \vee q$.

Finding the Truth Value: Disjunction, i.e., “or”

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Let p represent “I am a man” and q represent “I am an Elephant”.
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Finding the Truth Value: Disjunction, i.e., “or”

Example 1.3.4.

Let p represent “I am a man” and q represent “I am an Elephant”.
Find the truth value of $p \vee q$.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	
F	F	

- Determine which statements are true and false, and choose the corresponding answer.

Finding the Truth Value: Disjunction, i.e., “or”

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p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	

- Determine which statements are true and false, and choose the corresponding answer.

Finding the Truth Value: Disjunction, i.e., “or”

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p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- Determine which statements are true and false, and choose the corresponding answer.

Finding the Truth Value: Disjunction, i.e., “or”

Example 1.3.5.

Let p represent $0 > 3$, and let q represent $0 < 2$. Find the truth value of the compound statement $p \vee q$

Finding the Truth Value: Disjunction, i.e., “or”

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Let p represent $0 > 3$, and let q represent $0 < 2$. Find the truth value of the compound statement $p \vee q$

Using a *truth table*.

Finding the Truth Value: Disjunction, i.e., “or”

Example 1.3.5.

Let p represent $0 > 3$, and let q represent $0 < 2$. Find the truth value of the compound statement $p \vee q$

Using a *truth table*.

p	q	$p \vee q$

Compound Statements

Example 1.3.6.

Let p represent $0 > 3$, and let q represent $0 < 2$. Find the truth value of the compound statement $\sim (\sim p \wedge q)$

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p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

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T	T	T
T	F	F
F	T	F
F	F	F

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T	T	T
T	F	T
F	T	T
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p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$\sim p$	$\sim p \wedge q$	$\sim(\sim p \wedge q)$
T	T			
T	F			
F	T	T		
F	F			

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p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$\sim p$	$\sim p \wedge q$	$\sim(\sim p \wedge q)$
T	T			
T	F			
F	T	T	T	
F	F			

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p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$\sim p$	$\sim p \wedge q$	$\sim(\sim p \wedge q)$
T	T			
T	F			
F	T	T	T	F
F	F			

Compound Statements

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p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$$\sim(\sim p \wedge q)$$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
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p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$$\sim(\sim p \wedge q)$$

$$\sim(\sim F \wedge T)$$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
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Compound Statements

Example 1.3.6.

Let p represent $0 > 3$, and let q represent $0 < 2$. Find the truth value of the compound statement $\sim(\sim p \wedge q)$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$$\sim(\sim p \wedge q)$$

$$\sim(\sim F \wedge T)$$

$$\sim(T \wedge T)$$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Compound Statements

Example 1.3.6.

Let p represent $0 > 3$, and let q represent $0 < 2$. Find the truth value of the compound statement $\sim(\sim p \wedge q)$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$$\sim(\sim p \wedge q)$$

$$\sim(\sim F \wedge T)$$

$$\sim(T \wedge T)$$

$$\sim(T)$$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
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Compound Statements

Example 1.3.6.

Let p represent $0 > 3$, and let q represent $0 < 2$. Find the truth value of the compound statement $\sim (\sim p \wedge q)$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$$\sim (\sim p \wedge q)$$

$$\sim (\sim F \wedge T)$$

$$\sim (T \wedge T)$$

$$\sim (T)$$

F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Compound Statements

Example 1.3.7.

Let p represent a true statement, and let q and r represent false statements. Find the truth value of the compound statement

$$(p \wedge r) \vee \sim q$$

Compound Statements

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Let p represent a true statement, and let q and r represent false statements. Find the truth value of the compound statement

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Compound Statements

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Let p represent a true statement, and let q and r represent false statements. Find the truth value of the compound statement

$$(p \wedge r) \vee \sim q$$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Compound Statements

Example 1.3.7.

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$$(p \wedge r) \vee \sim q$$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$$(p \wedge r) \vee \sim q$$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Finding the Truth Value: Compound Statements

Example 1.3.8.

If q is true, what is the truth value of $q \vee (q \wedge \sim p)$

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p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

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T	T	T
T	F	T
F	T	T
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Finding the Truth Value: Compound Statements

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If q is true, what is the truth value of $q \vee (q \wedge \sim p)$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
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$$q \vee (q \wedge \sim p)$$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Your Turn: Find the Truth Value

Problem 1.3.1. Let p represent a false statement and q represent a true statement. Find the truth value of each compound statement.

$$\sim p$$

Your Turn: Find the Truth Value

Problem 1.3.2. Let p represent a false statement and q represent a true statement. Find the truth value of each compound statement.

$$p \vee \sim q$$

Your Turn: Find the Truth Value

Problem 1.3.3. Let p represent a false statement and q represent a true statement. Find the truth value of each compound statement.

$$\sim (p \vee \sim q)$$

Your Turn: Find the Truth Value

Problem 1.3.4. Let p represent a true statement, and q and r represent false statements. Find the truth value of each compound statement.

$$(q \vee \sim r) \wedge p$$

Your Turn: Find the Truth Value

Problem 1.3.5. Let p represent a true statement, and q and r represent false statements. Find the truth value of each compound statement.

$$(\sim p \wedge q) \vee \sim r$$

Your Turn: Find the Truth Value

Problem 1.3.6. Let p be the statement " $2 > 7$ ", q be statement " $8 \leq 6$," and r be the statement " $19 \leq 19$." Find the truth value of each compound statement.

$$\sim p \vee \sim r$$

Your Turn: Find the Truth Value

Problem 1.3.7. Let p be the statement " $2 > 7$ ", q be statement " $8 \leq 6$," and r be the statement " $19 \leq 19$." Find the truth value of each compound statement.

$$(p \wedge q) \vee r$$

Your Turn: Find the Truth Value

Problem 1.3.8. Let p be the statement " $2 > 7$ ", q be statement " $8 \leq 6$," and r be the statement " $19 \leq 19$." Find the truth value of each compound statement.

$$(\sim r \wedge q) \vee \sim p$$

Truth Tables: Conditional, i.e., “if...then”

Example 1.3.9.

Construct a truth table for the statement “If you give me \$50 then I will give you a ticket”.

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Construct a truth table for the statement “If you give me \$50 then I will give you a ticket”.

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Truth Tables: Conditional, i.e., “if...then”

Example 1.3.9.

Construct a truth table for the statement “If you give me \$50 then I will give you a ticket”.

p : You gave me \$50.

q : I gave you a ticket.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	

Truth Tables: Conditional, i.e., “if...then”

Example 1.3.9.

Construct a truth table for the statement “If you give me \$50 then I will give you a ticket”.

p : You gave me \$50.

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T	T	T
T	F	F
F	T	T
F	F	T

Because $p \rightarrow q$ is not false

Truth Tables: Conditional, i.e., “if...then”

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Construct a truth table for the statement “If you give me \$50 then I will give you a ticket”.

p : You gave me \$50.

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Truth Tables: Conditional, i.e., “if...then”

Example 1.3.9.

Construct a truth table for the statement “If you give me \$50 then I will give you a ticket”.

p : You gave me \$50.

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p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Because $p \rightarrow q$ is not false

Because $p \rightarrow q$ is not false

Problem 1.3.9. Translate the compound statement into symbolic form and the construct the truth table for the expression.

If it is raining then the streets are wet.

Problem 1.3.10. Translate the compound statement into symbolic form and then construct the truth table for the expression.

The water supply is rationed if it does not rain.

Problem 1.3.11. Translate the compound statement into symbolic form and then construct the truth table for the expression.

All triangles are polygons.

Problem 1.3.12. Translate the compound statement into symbolic form and the construct the truth table for the expression.

No hexagon is a triangle.

Problem 1.3.13. Translate the compound statement into symbolic form and then construct the truth table for the expression.

Being a triangle is sufficient for not being a square.

Problem 1.3.14. Translate the compound statement into symbolic form and the construct the truth table for the expression.

Not being a triangle is necessary for being a square.

Problem 1.3.15. Translate the compound statement into symbolic form and then construct the truth table for the expression.

If he does not go to jail, then he is innocent or has an alibi.

Problem 1.3.16. Translate the compound statement into symbolic form and the construct the truth table for the expression.

If proposition A passes and Proposition B does not, jobs are lost or taxes will be raised.

Example 1.3.10.

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- 1 The streets are wet, or it is not raining.

Equivalent Statements

Example 1.3.10.

- 1 The streets are wet, or it is not raining. $p \vee \sim q$
- 2 If it is raining, then the streets are wet.

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- 1 The streets are wet, or it is not raining. $p \vee \sim q$
- 2 If it is raining, then the streets are wet. $p \rightarrow q$

Equivalent Statements

Example 1.3.10.

- 1 The streets are wet, or it is not raining. $p \vee \sim q$
- 2 If it is raining, then the streets are wet. $p \rightarrow q$

p : The streets are wet

q : it is raining

Equivalent Statements

Example 1.3.10.

- ① The streets are wet, or it is not raining. $p \vee \sim q$
- ② If it is raining, then the streets are wet. $p \rightarrow q$

p : The streets are wet

q : it is raining

Equivalent Statements

Example 1.3.10.

- 1 The streets are wet, or it is not raining. $p \vee \sim q$
- 2 If it is raining, then the streets are wet. $p \rightarrow q$

p : The streets are wet

q : it is raining

		1	2
p	q	$p \vee \sim q$	$q \rightarrow p$
T	T	T	
T	F	T	
F	T	F	
F	F		

Equivalent Statements

Example 1.3.10.

- ① The streets are wet, or it is not raining. $p \vee \sim q$
- ② If it is raining, then the streets are wet. $p \rightarrow q$

p : The streets are wet

q : it is raining

		①	②
p	q	$p \vee \sim q$	$q \rightarrow p$
T	T	T	
T	F	T	
F	T	F	
F	F	T	

Equivalent Statements

Example 1.3.10.

- 1 The streets are wet, or it is not raining. $p \vee \sim q$
- 2 If it is raining, then the streets are wet. $p \rightarrow q$

p : The streets are wet

q : it is raining

p	q	¹ $p \vee \sim q$	² $q \rightarrow p$
T	T	T	T
T	F	T	
F	T	F	
F	F	T	

Equivalent Statements

Example 1.3.10.

- ① The streets are wet, or it is not raining. $p \vee \sim q$
- ② If it is raining, then the streets are wet. $p \rightarrow q$

p : The streets are wet

q : it is raining

		①	②
p	q	$p \vee \sim q$	$q \rightarrow p$
T	T	T	T
T	F	T	T
F	T	F	F
F	F	T	T

Equivalent Statements

Example 1.3.10.

- 1 The streets are wet, or it is not raining. $p \vee \sim q$
- 2 If it is raining, then the streets are wet. $p \rightarrow q$

p : The streets are wet

q : it is raining

		1	2
p	q	$p \vee \sim q$	$q \rightarrow p$
T	T	T	T
T	F	T	T
F	T	F	F
F	F	T	T

Equivalent Statements

Example 1.3.10.

- 1 The streets are wet, or it is not raining. $p \vee \sim q$
- 2 If it is raining, then the streets are wet. $p \rightarrow q$

p : The streets are wet

q : it is raining

		1	2
p	q	$p \vee \sim q$	$q \rightarrow p$
T	T	T	T
T	F	T	T
F	T	F	F
F	F	T	T

We say that $p \vee \sim q$ and $q \rightarrow p$ are *equivalent expressions*, written $p \vee \sim q \equiv q \rightarrow p$.

Equivalent expressions

Equivalence

Symbolic expressions that have identical truth values are called *equivalent expressions*, notated using " $\equiv$ ".

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Symbolic expressions that have identical truth values are called *equivalent expressions*, notated using “ $\equiv$ ”.

ex. $p \vee \sim q \equiv p \rightarrow q$

ex. "It is raining"

"It is not the case, that it is not raining"

Equivalent expressions

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Symbolic expressions that have identical truth values are called *equivalent expressions*, notated using “ $\equiv$ ”.

ex. $p \vee \sim q \equiv p \rightarrow q$

ex. "It is raining"

"It is not the case, that it is not raining"

$$q \equiv \sim(\sim q)$$

ex. "If the streets are wet then it is raining"

"If it is not raining then the streets are not wet."

Equivalent expressions

Equivalence

Symbolic expressions that have identical truth values are called *equivalent expressions*, notated using “ $\equiv$ ”.

ex. $p \vee \sim q \equiv p \rightarrow q$

ex. "It is raining"

"It is not the case, that it is not raining"

$$q \equiv \sim(\sim q)$$

ex. "If the streets are wet then it is raining"

"If it is not raining then the streets are not wet."

$$q \rightarrow p \equiv \sim p \rightarrow \sim q$$

Problem 1.3.17. Determine whether the two statements are equivalent.

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- She is unemployed or she does not have a high school diploma.

Problem 1.3.17. Determine whether the two statements are equivalent.

- She is unemployed or she does not have a high school diploma.
- If she is employed then she does not have a high school diploma.

Problem 1.3.17. Determine whether the two statements are equivalent.

- She is unemployed or she does not have a high school diploma.
- If she is employed then she does not have a high school diploma.

Equivalent expressions

Problem 1.3.17. Determine whether the two statements are equivalent.

- She is unemployed or she does not have a high school diploma.
- If she is employed then she does not have a high school diploma.

p : She is employed.

q : She has a HS diploma.

Equivalent expressions

Problem 1.3.17. Determine whether the two statements are equivalent.

- She is unemployed or she does not have a high school diploma.
- If she is employed then she does not have a high school diploma.

p : She is employed.

q : She has a HS diploma.

Equivalent expressions

Problem 1.3.17. Determine whether the two statements are equivalent.

- She is unemployed or she does not have a high school diploma.
- If she is employed then she does not have a high school diploma.

p : She is employed.

q : She has a HS diploma.

p	q		
T	T		
T	F		
F	T		
F	F		

Problem 1.3.18. Determine whether the two statements are equivalent.

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- If the spotted owl is on the endangered list, then lumber jobs are lost.

Problem 1.3.18. Determine whether the two statements are equivalent.

- If the spotted owl is on the endangered list, then lumber jobs are lost.
- If lumber jobs are not lost, then the spotted owl is not on the endangered list.

Problem 1.3.18. Determine whether the two statements are equivalent.

- If the spotted owl is on the endangered list, then lumber jobs are lost.
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Equivalent expressions

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- If the spotted owl is on the endangered list, then lumber jobs are lost.
- If lumber jobs are not lost, then the spotted owl is not on the endangered list.

p :

q :

Equivalent expressions

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q :

Equivalent expressions

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- If lumber jobs are not lost, then the spotted owl is not on the endangered list.

p :

q :

p	q		
T	T		
T	F		
F	T		
F	F		

De Morgan's Law

De Morgan's Law

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$$\sim (p \wedge q) \equiv \sim p \vee \sim q$$

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$$\sim (p \wedge q) \equiv \sim p \vee \sim q$$

$$\sim (p \vee q) \equiv \sim p \wedge \sim q$$

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$$\sim (p \wedge q) \equiv \sim p \vee \sim q$$

$$\sim (p \vee q) \equiv \sim p \wedge \sim q$$

Example 1.3.11.

Negate the following statements

I have a college degree, and I am not employed.

De Morgan's Law

De Morgan's Law

$$\sim (p \wedge q) \equiv \sim p \vee \sim q$$

$$\sim (p \vee q) \equiv \sim p \wedge \sim q$$

Example 1.3.11.

Negate the following statements

I have a college degree, and I am not employed.

p : I have a college degree.

q : I am not employed.

De Morgan's Law

De Morgan's Law

$$\sim (p \wedge q) \equiv \sim p \vee \sim q$$

$$\sim (p \vee q) \equiv \sim p \wedge \sim q$$

Example 1.3.11.

Negate the following statements

I have a college degree, and I am not employed.

p : I have a college degree.

q : I am not employed.

Negate, $p \wedge q$.

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Negate, $p \wedge q$.

$$\sim (p \wedge q)$$

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Negate, $p \wedge q$.

$$\sim (p \wedge q)$$

$$\equiv \sim p \vee \sim q$$

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p : I have a college degree.

q : I am not employed.

Negate, $p \wedge q$.

$$\sim (p \wedge q)$$

$$\equiv \sim p \vee \sim q$$

$$\equiv \sim p \vee \sim q$$

De Morgan's Law

Negation

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T		
T	F		
F	T		
F	F		

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	
T	F		
F	T		
F	F		

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	
T	F	F	
F	T		
F	F		

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	
T	F	F	
F	T	T	
F	F		

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	
T	F	F	
F	T	T	
F	F	T	

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	T
T	F	F	
F	T	T	
F	F	T	

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	T
T	F	F	F
F	T	T	
F	F	T	

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

So, $p \rightarrow q \equiv \sim p \vee q$.

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

So, $p \rightarrow q \equiv \sim p \vee q$.

So then $\sim (p \rightarrow q) \equiv \sim (\sim p \vee q)$.

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

So, $p \rightarrow q \equiv \sim p \vee q$.

So then $\sim (p \rightarrow q) \equiv \sim (\sim p \vee q)$.

By De Morgan's Law,

$\sim (\sim p \vee q) \equiv \sim (\sim p) \wedge \sim q \equiv$
 $p \wedge \sim q$.

De Morgan's Law

Negation

Negation of *negation*: $\sim (\sim p) \equiv p$

Negation of *conjunction*: $\sim (p \wedge q) \equiv \sim p \vee \sim q$

Negation of *disjunction*: $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Negation of *conditional*: $\sim (p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \rightarrow q$	$\sim p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

So, $p \rightarrow q \equiv \sim p \vee q$.

So then $\sim (p \rightarrow q) \equiv \sim (\sim p \vee q)$.

By De Morgan's Law,

$$\sim (\sim p \vee q) \equiv \sim (\sim p) \wedge \sim q \equiv p \wedge \sim q.$$
$$\therefore \sim (p \rightarrow q) \equiv p \wedge \sim q$$

Problems

Problem 1.3.19. Write the statement in symbolic form, and find the symbolic negation of the statement.

The TV is broken, or there is a power outage.

Problem 1.3.20. Write the statement in symbolic form, and find the symbolic negation of the statement.

If the building contains asbestos, then the original contractor is responsible.

Problem 1.3.21. Determine which statements are equivalent

1. Knowing Morse code is sufficient for operating a telegraph.
2. Knowing Morse code is necessary for operating a telegraph.

Problem 1.3.22. Translate the compound statement into symbolic form and the construct the truth table for the expression.

If proposition A passes and Proposition B does not, jobs are lost or taxes will be raised.